

Skills Summary

Solving Trigonometric Equations on a Restricted Interval

Use this reference while completing your worksheet or when studying.

Skill 1 — Linear equations in one ratio

Method: isolate the ratio, find the reference angle from the positive value, then place one solution in *each* quadrant the sign allows.

Sine positive: I, II. Sine negative: III, IV. Cosine positive: I, IV. Cosine negative: II, III. Tangent positive: I, III. Tangent negative: II, IV.

Example: Solve $2 \sin \theta - \sqrt{3} = 0$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. One ratio, first power — so isolate it and read off the two quadrants where that sign occurs.

Step 2. $\sin \theta = \frac{\sqrt{3}}{2}$: reference angle 60° , sine positive in I and II. $\Rightarrow \theta = 60^\circ$ or $\theta = 120^\circ$.

Try it: Solve $2 \cos \theta + \sqrt{2} = 0$ for $0^\circ \leq \theta < 360^\circ$.

check: 135° or $225^\circ = \theta$

Skill 2 — Equations that factor

Rule: a product is zero exactly when one of its factors is zero. Set each factor to zero and solve it as a separate linear equation, then merge the lists.

Never divide by a trigonometric factor — dividing by $\cos \theta$ throws away every solution with $\cos \theta = 0$.

Example: Solve $\cos \theta (2 \cos \theta + 1) = 0$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. A product equals zero, so split it into two easier equations rather than expanding.

Step 2. $\cos \theta = 0 \Rightarrow 90^\circ, 270^\circ$; $\cos \theta = -\frac{1}{2} \Rightarrow 120^\circ, 240^\circ$. $\Rightarrow \theta = 90^\circ, 120^\circ, 240^\circ, 270^\circ$.

Try it: Solve $\sin \theta (2 \sin \theta + 1) = 0$ for $0^\circ \leq \theta < 360^\circ$.

check: $0^\circ, 180^\circ, 210^\circ, 330^\circ = \theta$

Skill 3 — Identity first, then factor

When two different ratios appear, use $\sin^2 \theta + \cos^2 \theta = 1$ to rewrite the squared one, leaving a quadratic in a single ratio. Then factor and use Skill 2.

$$\sin^2 \theta = 1 - \cos^2 \theta \quad \cos^2 \theta = 1 - \sin^2 \theta$$

Example: Solve $2 \sin^2 \theta + \cos \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. Two ratios in one equation cannot be solved directly, so trade the squared sine for cosine and get a quadratic in cosine alone.

Step 2. $2(1 - \cos^2 \theta) + \cos \theta - 1 = -(2 \cos \theta + 1)(\cos \theta - 1)$, so $\cos \theta = -\frac{1}{2}$ or $\cos \theta = 1$.
 $\Rightarrow \theta = 0^\circ, 120^\circ, 240^\circ$.

Try it: Solve $2 \cos^2 \theta + \sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

check: $\theta = 0^\circ, 120^\circ, 240^\circ$

Skill 4 — Multiple angles: widen first

For $\sin(k\theta)$ or $\cos(k\theta)$: substitute $u = k\theta$ and multiply the interval's endpoints by k . Solve for u over that wider interval, then divide every answer by k .

Expect about k times as many solutions — if you get the same number as a plain equation, you widened too late.

Example: Solve $2 \cos 2\theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. The angle inside is doubled, so the inside variable sweeps twice as far — widen the interval before solving, not after.

Step 2. $u = 2\theta$ on $[0^\circ, 720^\circ)$, $\cos u = \frac{1}{2} \Rightarrow u = 60^\circ, 300^\circ, 420^\circ, 660^\circ$; halve each. $\Rightarrow \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$.

Try it: Solve $2 \sin 3\theta - 1 = 0$ for $0^\circ \leq \theta < 180^\circ$.

check: $\theta = 10^\circ, 50^\circ, 130^\circ, 170^\circ$